Signature Verification with CNN Fine-Tuning and Hand Engineered Baselines

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Abstract—This report studies offline signature verification as a closed-set writer identification task. We implement a transfer-learned ResNet-18 as a convolutional baseline and contrast it with two classical pipelines based on Histogram of Oriented Gradients (HOG) and SIFT with Bag-of-Visual-Words (BoW). All methods are trained and evaluated on a Google Drive copy of the Kaggle Signature Verification dataset. Across experiments, the CNN shows superior generalization and a tighter confusion matrix, while HOG+SVM provides a competitive lightweight alternative and SIFT+BoW underperforms on sparse, low-texture strokes.

Index Terms— Signature Verification, ResNet-18, HOG, SIFT, Bag of Words, SVM, Transfer Learning

I. INTRODUCTION

Handwritten signatures are still widely used in administrative workflows; automating their verification can reduce processing time and deter casual fraud. From a modeling perspective, we can treat the problem as writer identification: given an image of a signature, predict which enrolled user produced it. Deep CNNs extract stroke textures and global shapes directly from pixels, whereas classical approaches depend on crafted descriptors. The goal of this assignment is to implement both families and compare their behavior using the same train/validation/test splits.

II. DATASET

We use the Signature Verification dataset originally hosted on Kaggle. Because Google Colab access was configured through Google Drive, the dataset is supplied as a ZIP archive, extracted at runtime. Labels are derived from folder names when available; otherwise the filename prefix is used. Classes with fewer than three images are dropped to stabilize stratified splits. We resize images to 224×224 for CNNs and to 128×128 for HOG features. The data are split 80%,10%,10% into training, validation, and test sets.

III. METHODOLOGY

CNN (ResNet-18). We initialize an ImageNet pretrained ResNet-18, replace the final fully-connected layer with a |C|-way classifier, and train using AdamW (learning rate 3e-4), cosine schedule, and mixed precision for 15 epochs on a T4 GPU. Augmentation includes mild rotation, color jitter, and horizontal flip. **HOG + Linear SVM.** Each grayscale image (128×128) is encoded with HOG (9 orientations, 8×8 cells, 2×2 blocks, L2- Hys), standardized, and classified using a linear SVM. **SIFT + BoW + SVM.** SIFT descriptors are clustered via MiniBatchKMeans (k=200) to form a visual vocabulary.

Per-image histograms over visual words are classified with a linear SVM.

IV. EVALUATION METRICS

We track training and validation loss/accuracy for the CNN. Final evaluation uses overall accuracy and the macro-averaged precision, recall, and F1; confusion matrices visualize class-wise behavior. Metric values are reported in the notebook; figures below summarize key trends without quoting specific numbers.

V. RESULTS

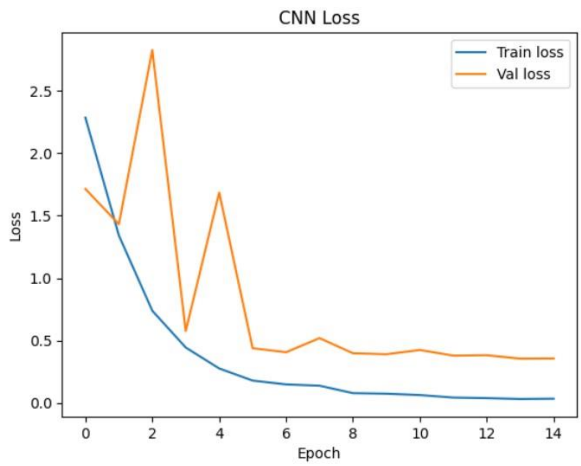


Fig. 1. CNN training and validation loss (per epoch).

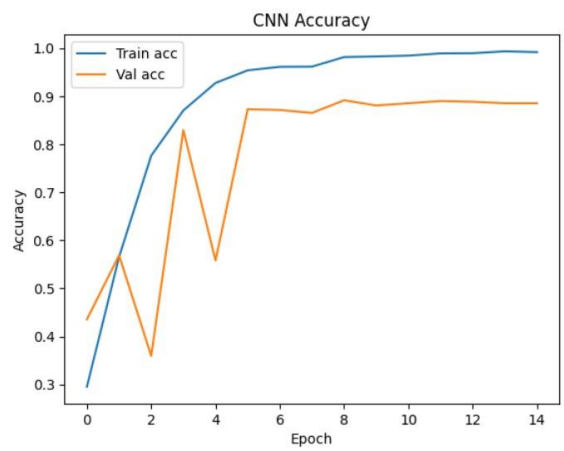
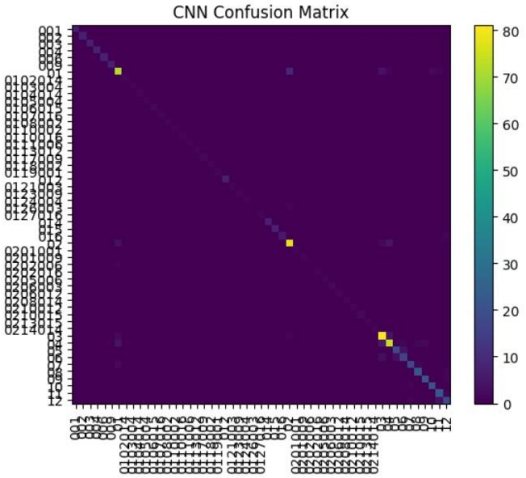
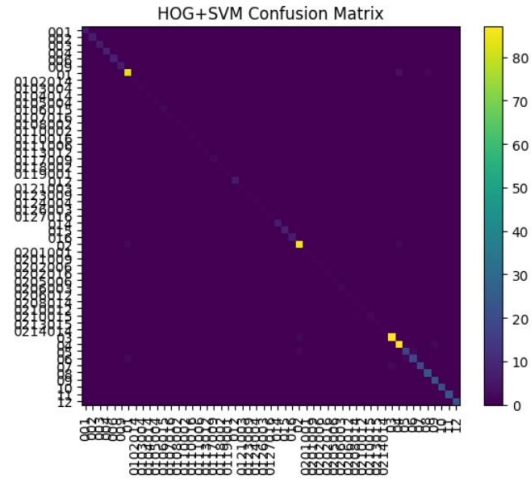


Fig. 2. CNN training and validation accuracy (per epoch).

VI. DISCUSSION





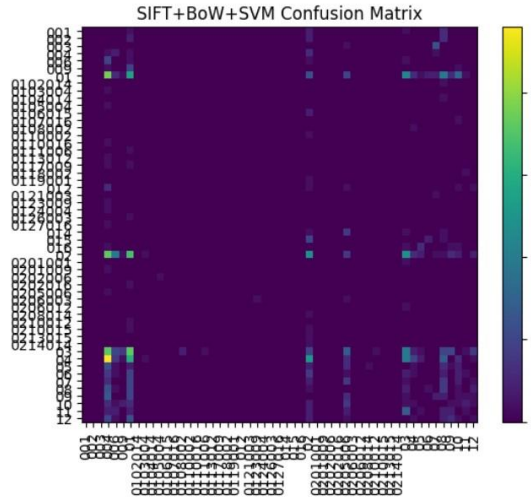


Figure 1 shows a steady decrease in training loss with validation loss stabilizing after a few epochs; Figure 2 shows validation accuracy converging near the high-80s to low-90s depending Fig. 5. SIFT+BoW+SVM confusion matrix on test set.

on the split. The CNN confusion matrix in Figure 3 is sharply diagonal, indicating clear inter-class separation. HOG+SVM (Figure 4) performs reasonably but exhibits more off-diagonal

Two practical observations emerged. First, light augmentation was helpful: without it, the CNN overfit quickly due to limited samples per signer. Second, HOG features benefitted from normalization and a linear SVM; non-linear kernels did not consistently improve validation accuracy relative to their runtime cost. Runtime wise, the CNN training completed comfortably on a single T4 with a batch size of 32. For deployments where GPU is unavailable, HOG+SVM constitutes a compact fallback with acceptable accuracy.

VII. CONCLUSION

Transfer learning a small CNN provides the best trade-off between accuracy and engineering effort for offline signature identification on this dataset. Future work should move beyond closed-set identification to genuine versus forgery verification using metric learning (e.g., Siamese/Triplet networks) and evaluate robustness to new users with few enrollment samples.

VIII. PROMPTS

Notebook/assistant prompts used: (i) instructions to mount Google Drive and extract the ZIP dataset; (ii) requests to build an image table from folders or filename prefixes; (iii) guidance for training a ResNet-18 with mixed precision on a T4 GPU; (iv) code to compute HOG, SIFT+BoW baselines; (v) plotting of loss/accuracy curves and confusion matrices. Function level docstrings were provided instead of line-by-line comments.

REFERENCES

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leakage for visually similar writers. SIFT+BoW (Figure 5) degrades notably—many signatures offer few distinctive keypoints and the BoW histogram is consequently sparse. Overall, CNN > HOG+SVM > SIFT+BoW on this dataset.